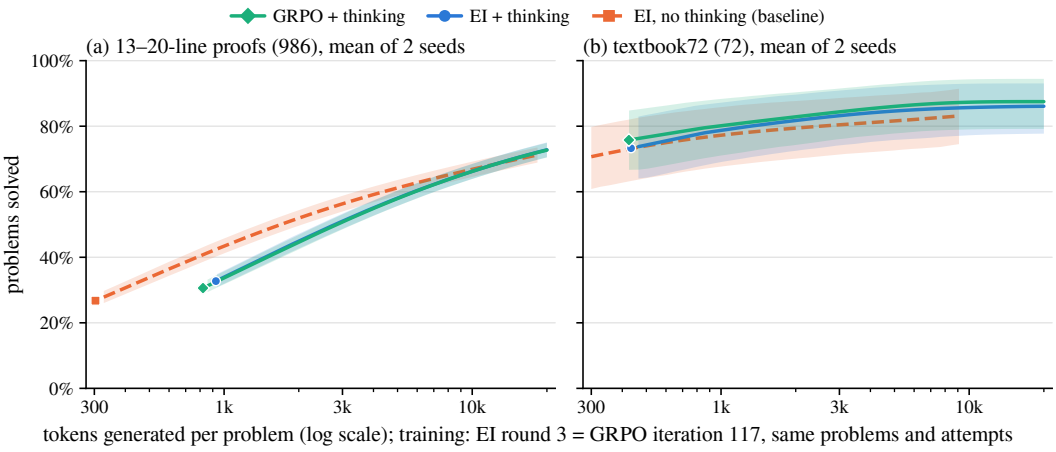
GRPO + thinking
EI + thinking
EI, no thinking (baseline)
(a) 13–20-line proofs (986), mean of 2 seeds
(b) textbook72 (72), mean of 2 seeds
problems solved
100%
80%
60%
40%
20%
0%
300
1k
3k
10k
tokens generated per problem (log scale); training: EI round 3 = GRPO iteration 117, same problems and attempts